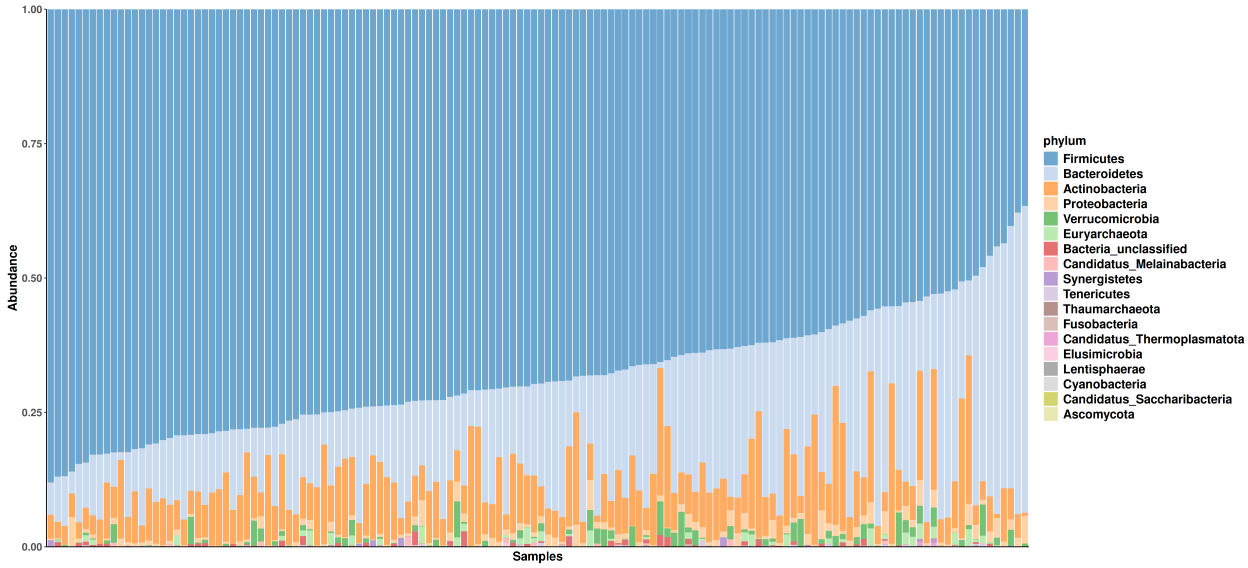
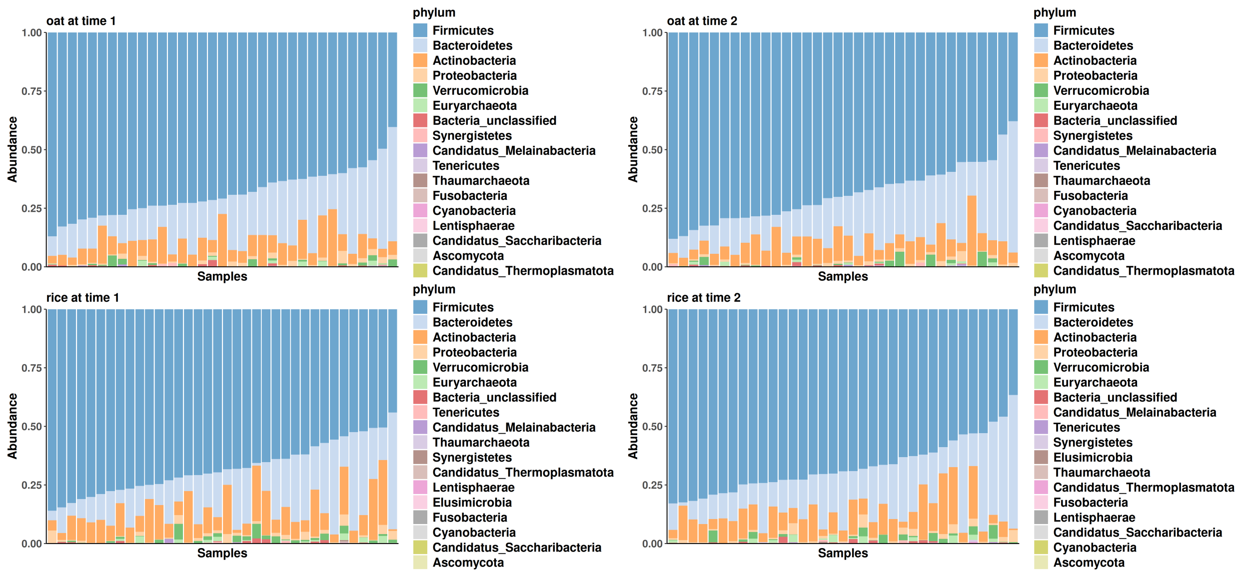
# Beta diversity

Muluh

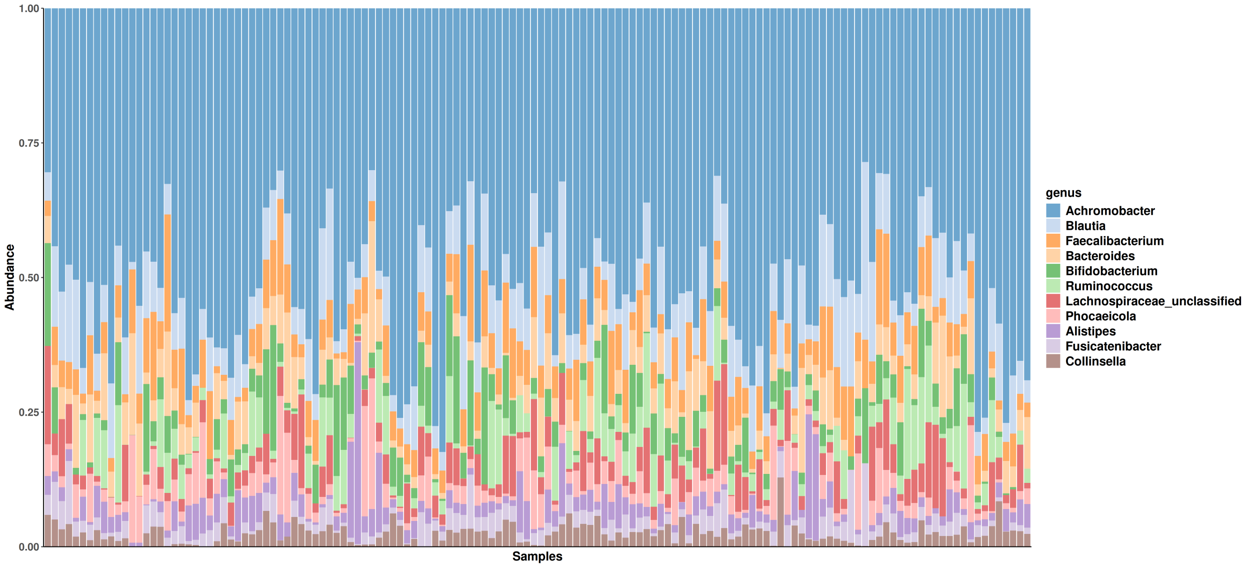
# 1 Community composition

## 1.1 Phylum

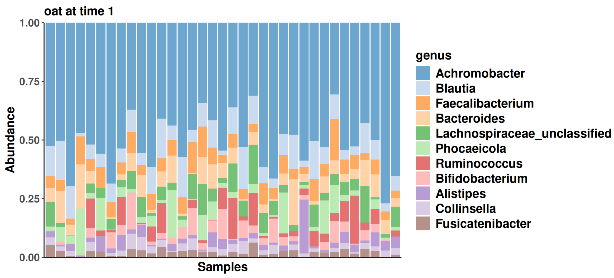


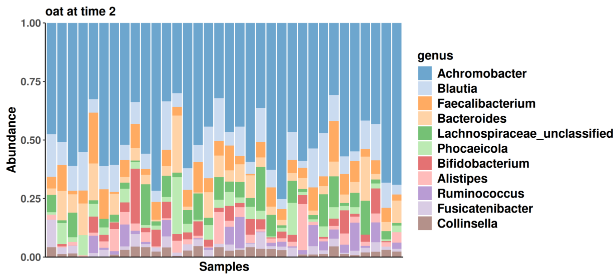
## 1.2 Genus



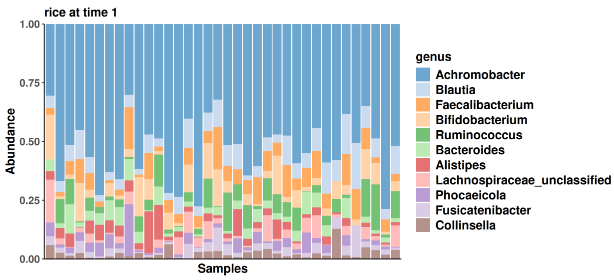
1.00
0.75
0.50
0.25
0.00
Abundance
Samples
genus
Achromobacter
Blautia
Faecalibacterium
Bacteroides
Bifidobacterium
Ruminococcus
Lachnospiraceae_unclassified
Phocaeicola
Alistipes
Fusicatenibacter
Collinsella

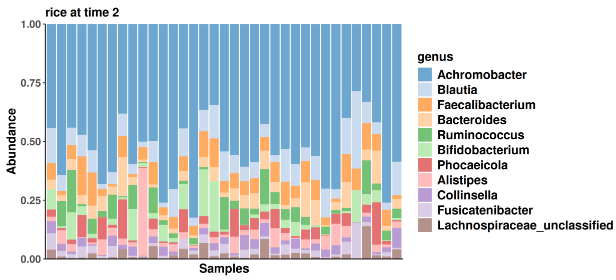
oat at time 1
1.00
0.75
0.50
0.25
0.00
Abundance
Samples
genus
Achromobacter
Blautia
Faecalibacterium
Bacteroides
Lachnospiraceae_unclassified
Phocaeicola
Ruminococcus
Bifidobacterium
Alistipes
Collinsella
Fusicatenibacter

oat at time 2
1.00
0.75
0.50
0.25
0.00
Abundance
Samples
genus
Achromobacter
Blautia
Faecalibacterium
Bacteroides
Lachnospiraceae_unclassified
Phocaeicola
Bifidobacterium
Alistipes
Ruminococcus
Fusicatenibacter
Collinsella

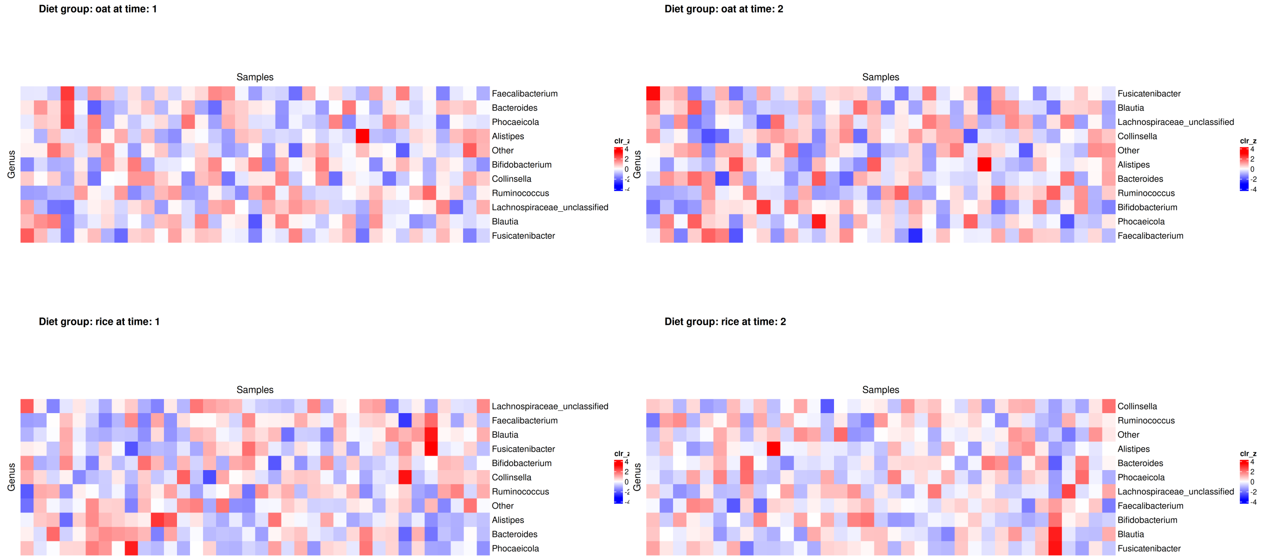



rice at time 1
1.00
0.75
0.50
0.25
0.00
Abundance
Samples
genus
Achromobacter
Blautia
Faecalibacterium
Bifidobacterium
Ruminococcus
Bacteroides
Alistipes
Lachnospiraceae_unclassified
Phocaeicola
Fusicatenibacter
Collinsella

rice at time 2
1.00
0.75
0.50
0.25
0.00
Abundance
Samples
genus
Achromobacter
Blautia
Faecalibacterium
Bacteroides
Ruminococcus
Bifidobacterium
Phocaeicola
Alistipes
Collinsella
Fusicatenibacter
Lachnospiraceae_unclassified

### 1.3 Heatmap



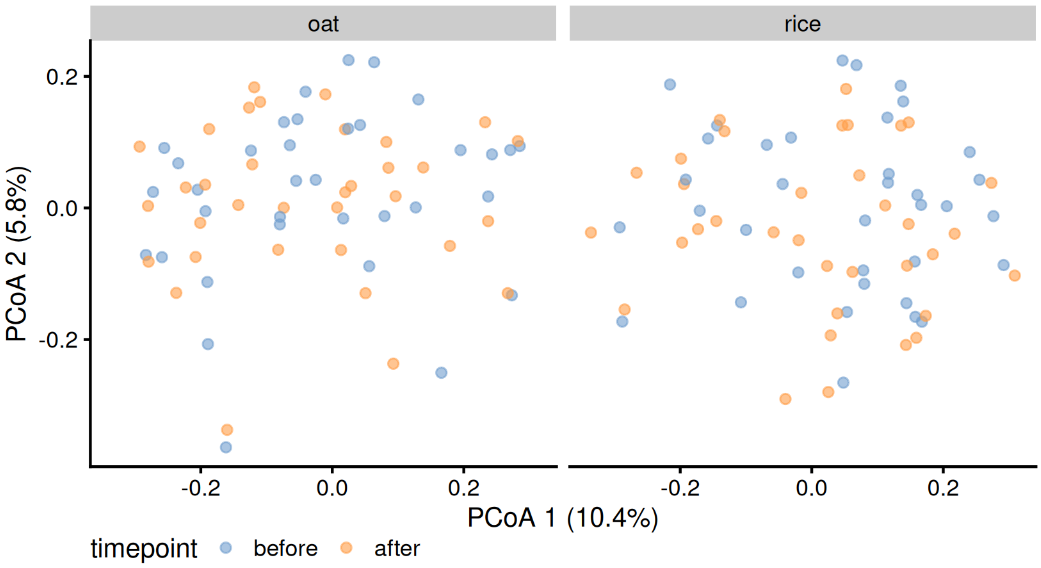

## 2 Unsupervised analysis

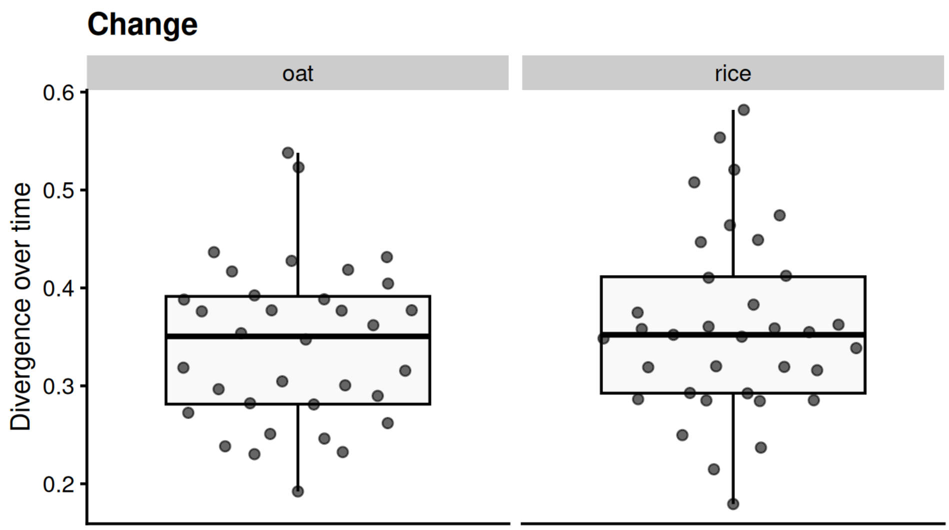
### 2.1 Community dissimilarity

This document presents the results of a microbiota project, focusing on beta diversity analysis. Specifically, in this section, we employed Principal Coordinates Analysis (PCoA) to assess community dissimilarity among microbiota samples.

- **Taxonomic level:** species
- **Dissimilarity index:** bray
- **Group variable:** diet
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon

## 2.2 PCoA plots




## 2.3 Table of the top findings

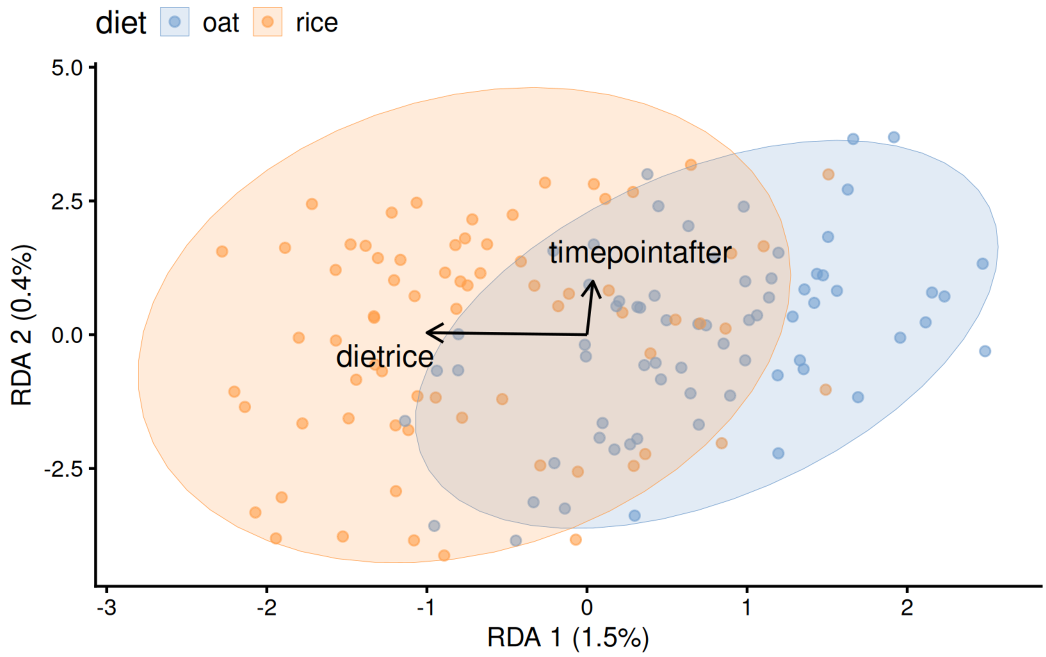
#### 2.3.0.1 Paired treatment between timepoint(1 and 2)

- Multiple testing correction: **paired fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of divergence paired between timepoint(1 and 2)

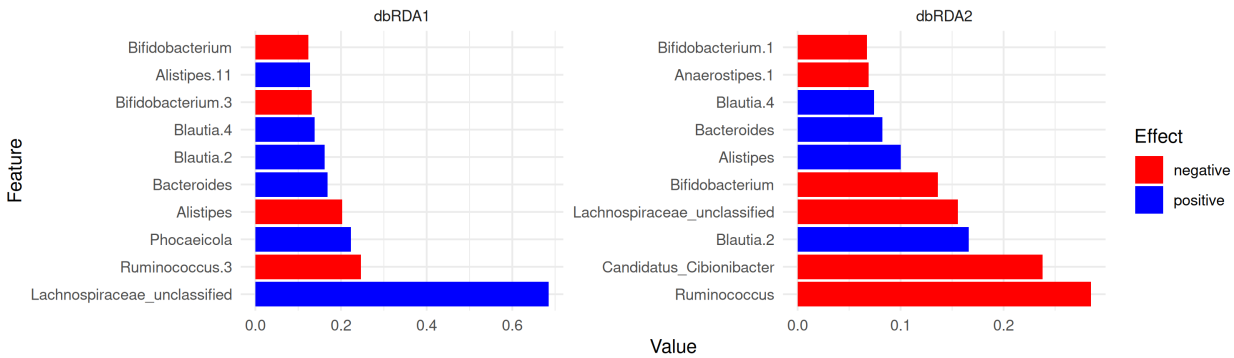
| Meal | mean | p.value | p.adj |
|---|---|---|---|
| oat | 0.3426224 | 4e-07 | 4e-07 |
| rice | 0.3612592 | 3e-07 | 4e-07 |

## 2.4 Significant variables




## 2.5 Significant taxa

- RDA




- MDS

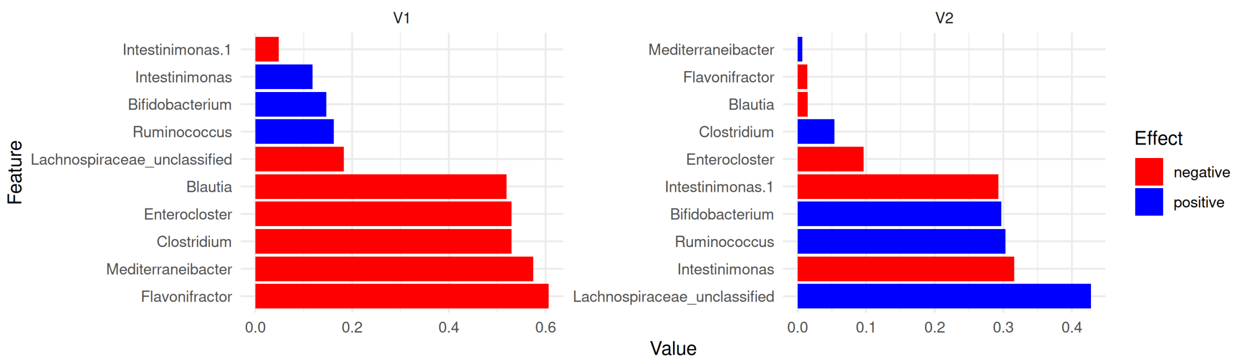


Table 2: PERMANOVA Results - before treatment

| **Group1** | **Group2** | **P_value** | **Homogeneity** | **p.adj** |
|---|---|---|---|---|
| oat | rice | 0.001 | 0.837 | 0.001 |

Table 3: PERMANOVA Results - after treatment

| **Group1** | **Group2** | **P_value** | **Homogeneity** | **p.adj** |
|---|---|---|---|---|
| oat | rice | 0.001 | 0.842 | 0.001 |

# 3 Supervised analysis

## 3.1 Permanova analysis

- Dissimilarity index: **bray**
- Group variable: **diet**

**Interpretation**

- Homogeneity $<$ 0.05: The assumption of homogeneity of variance is violated (i.e., the variances are significantly different across groups).
- Homogeneity  0.05: The assumption of homogeneity of variance is not violated (i.e., the variances are similar across groups).

### 3.1.1 Across-group

#### 3.1.1.1 After treatment

Table 4: PERMANOVA Results - Paired between timepoint (1 and 2)

| **Diet** | **P_value** | **Homogeneity** | **p.adj** |
|---|---|---|---|
| oat | 1.000 | 0.646 | 1 |
| rice | 0.858 | 0.817 | 1 |

#### 3.1.2 Within-group

# 4 Alternate analysis

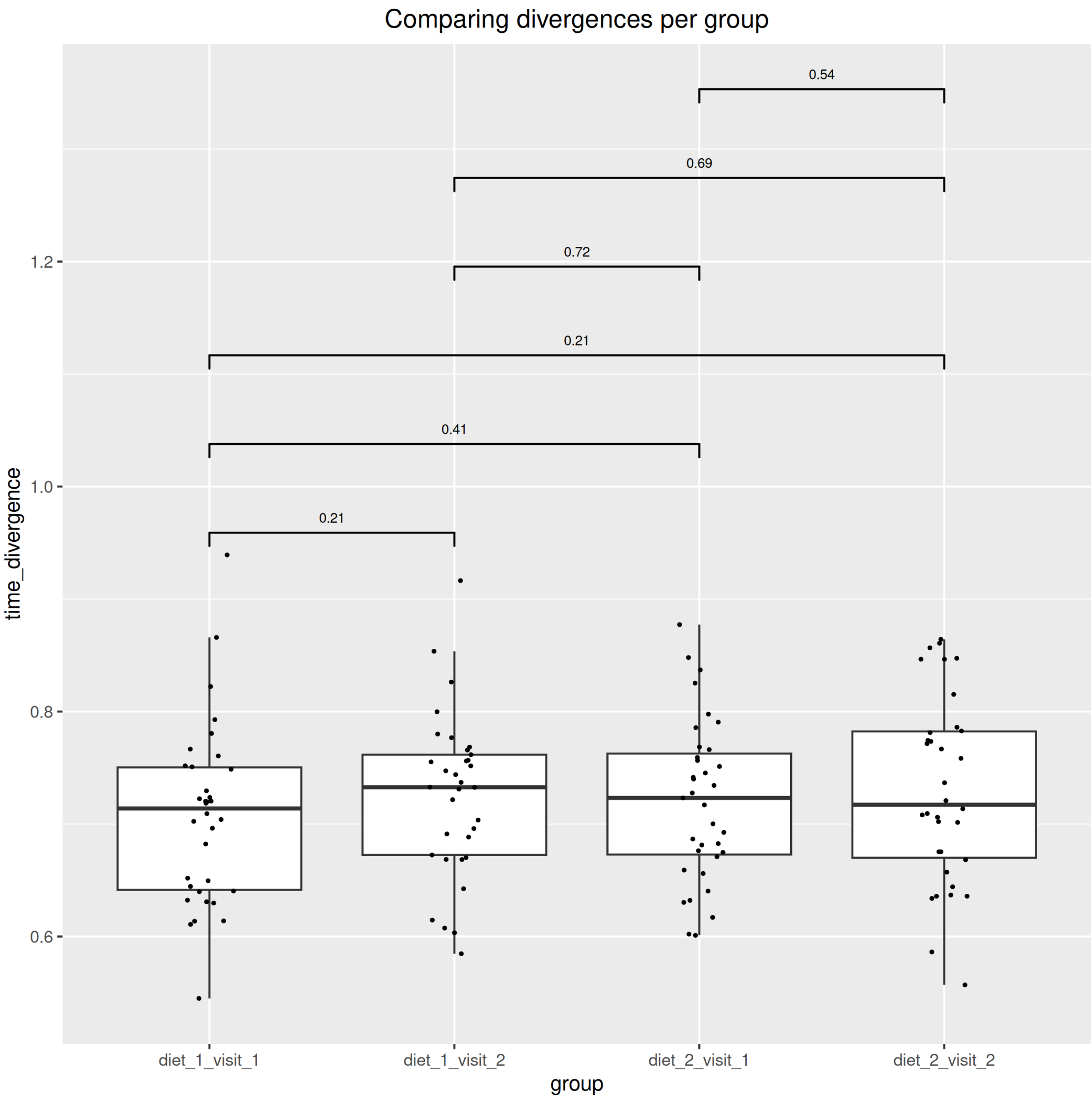



Table 5: PERMANOVA

| | before | after | change-1 | change-2 | change-3 | change-4 |
|---|---|---|---|---|---|---|
| betadisp_pval | 0.876 | 0.931 | NA | NA | NA | NA |
| permanova_pval | 0.333 | 0.105 | 0.999 | 0.976 | 0.997 | 1 |

## 4.1 Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
```

```
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(17): sample id ... time_difference time_divergence
reducedDimNames(1): PCoA_BC
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```